

Aim

To observe refraction and lateral deviation of a beam of light incident on a glass slab.

Apparatus and Material Required

Glass slab (size 10x6 cm), drawing board, drawing pins, plane sheet of paper, pins, common scale, protractor, pencil, rubber, cello tape, a sharply pointed pencil, a half meter scale.

Procedure

1. Fix a white paper sheet by drawing pins and drawing board.
2. Take a glass slab and put it symmetrically in the middle of the paper and mark its boundary ABCD.
3. Draw a normal at point Q on face AB and draw a line PQ making an angle i with the normal. PQ will represent an incident ray.
4. Fix two points 1 and 2 on the line PQ at distance of 10 cm or more between themselves.
5. See images of these two pins through face DC and fix two more pins at points 3 and 4 (10 cm or more apart) such that these two pins cover the images of first two pins, all being along a straight line.
6. Remove the glass slab. Draw straight line RS through points 3 and 4 to present emergent ray. Join QR to represent refracted ray.
7. Draw normal at point R on face DC and measure angle e . It comes to be equal to i .
8. Produce PQ forward to cut DC at N. Draw RK perpendicular to PN. RK measures lateral displacement at d .

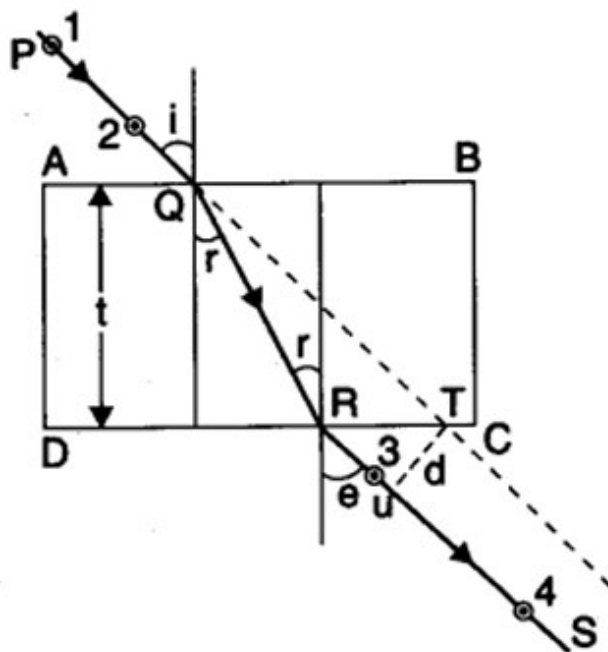
Observations

Angle of Incidence, i

Angle of refraction,

Thickness of glass slab, $t - AD = \dots\dots\dots$ cm

Lateral displacement, $d - TU = \dots\dots\dots$ cm



Calculations

Theoretical Value of displacement $d = t \sec r \sin(i-r)$ cm

Difference between measured and theoretical value of lateral displacement =cm

Conclusion / Result

1. The ray of light emitting out from a glass slab is parallel to the direction of incident ray, but is laterally deviated.
2. The angle of incident is equal to the angle of emergence, i.e. $i = e$.
3. The lateral deviation of emergent ray w.r.t. incident ray is directly proportional to the thickness of glass slab.

Precautions

1. The Upper surface of drawing board should be soft so that the pins can be penetrated easily.
2. The all pins must be vertically fixed on the board.
3. A sharp pencil should be used to draw boundary of the glass slab.
4. The glass slab should be of uniform thickness.
5. To locate the incident ray and emergent ray with greater accuracy, the distance between pins should be 6 cm to 8 cm.